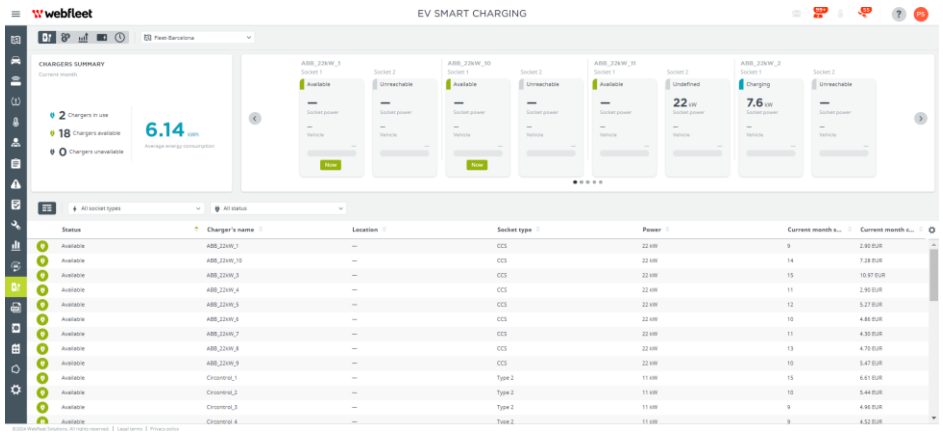

Webfleet EV Smart Charging

Charge better, save more

BRIDGESTONE
Solutions for your journey

Charge better, save more

Monitor and optimise all private chargers installed at your facility from Webfleet independent of their brand.

Webfleet EV Smart Charging is a new module providing two solutions: EV Charger Monitoring to monitor private chargers and EV Charger Optimisation to optimise charging routine helping fleets cut operational costs. Keep your EVs **charged up** and **ready to go**.

Content

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   - Requirements and price
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There are 5 key customer challenges while charging an EV

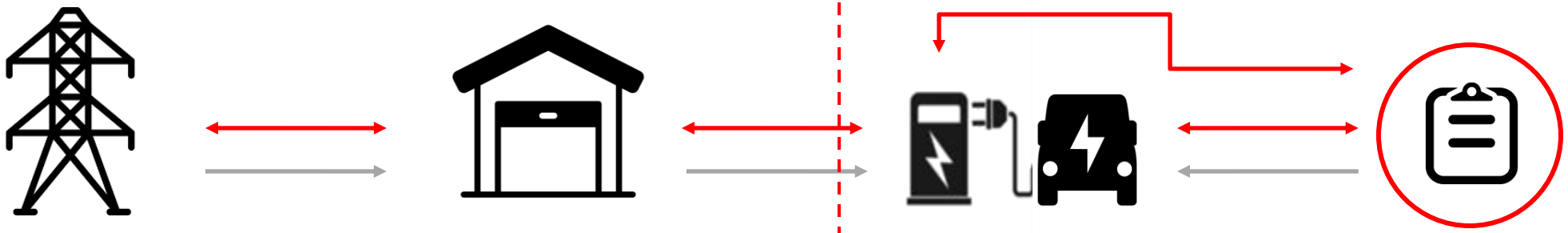
- "Where does your company charge its EVs today?"[1]
- **80%** said they charge at **private chargers** installed at **depots/warehouses**.
- What are the typical issues for businesses when using private charging?

		Impact to businesses			
	Planning	**High CAPEX**	**High charging costs**	**Lower vehicle residual value**	**Less CO_2 savings**
Issues	• Be able to easily check charger availability • Make sure vehicles are charged to run jobs	• Peak load consumption • Limited grid capacity • No optimisation on number of charging points	• Peak load consumption • Charging during peak tariff • No use of renewable energy to charge	• Battery degradation due to poor charging behavior (e.g. always charge up 100%)	• No use of renewable energy to charge

Smart Charging helps businesses reduce investments, cut costs, improve battery life and save CO_2

1 – Question asked to 700 business owners with electric vehicles on their fleets. Webfleet brand survey Q4/2022.

Smart charging balances energy demand and optimises your charging routine



Data flow		Energy grid	Warehouses	Private charger and vehicle	FMS/ Order management
		Transports energy to all warehouses	Get limited energy to power all operations	Charges vehicle for operation	Attributes jobs to vehicles
ON Smart Charging	↔	Lower risk of overload.	Balances consumption to use available capacity.	Charges only what is required for vehicle operational need	Prioritises charges and knows vehicle is ready
OFF Smart Charging	→	High risk of overload.	When consumption reaches a peak, higher charging cost or need to hire extra capacity.	Charges batteries up to pre-set value, irrespective of energy required for operations	Assumes vehicle is ready; high risk of vehicle service disruption

Webfleet EV Smart Charging in a nutshell

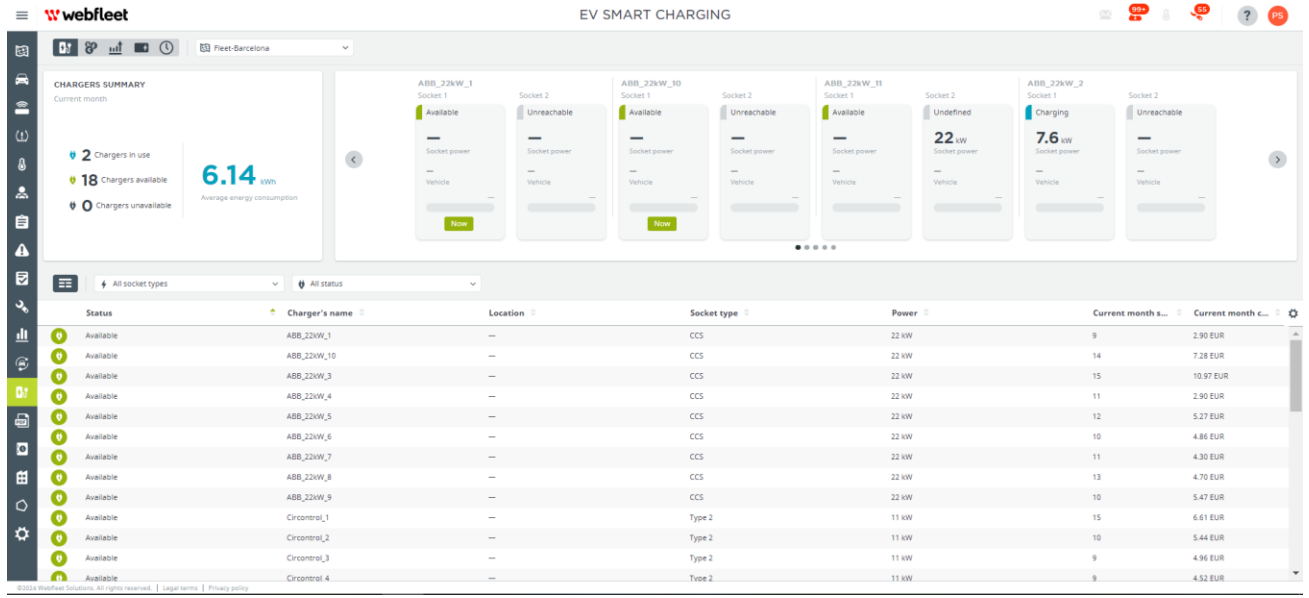
Webfleet EV Smart Charging is a new module in Webfleet designed for **businesses with private chargers[1] installed at company facilities. The service is composed of two solutions:**

EV Charger Monitoring

- Monitor charge points at individual and aggregate level.
- Consolidate all chargers in one place with Webfleet EV Smart Charging - no more juggling across platforms. Charger brand agnostic for seamless monitoring.
- Obtain insights on battery level and remaining charging time when plugged into the chargers[2].

EV Charger Optimisation

- Reduce costs, improve battery lifetime via our optimisation algorithm.
- Optimise charging routine with vehicle data improving fleet reliability.
- Prioritise charging using departure time.



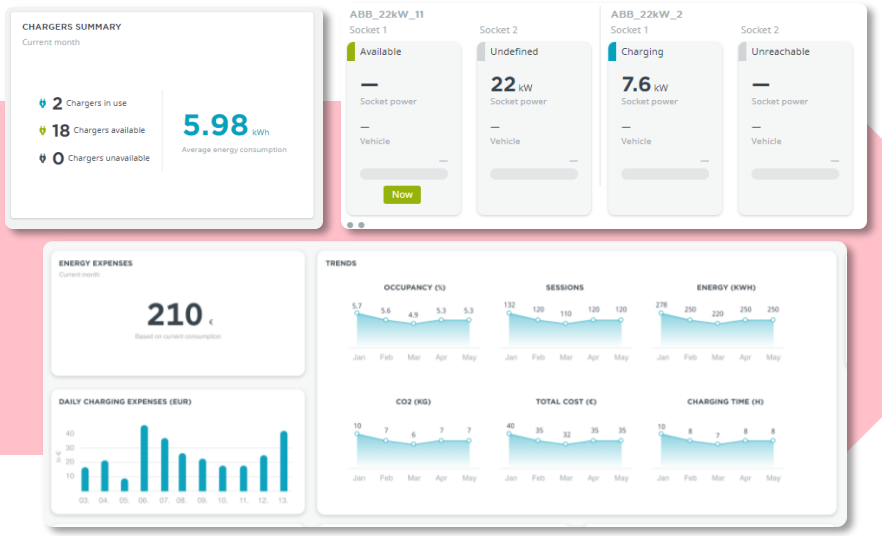
1 – Chargers are connected without extra hardware. They should support OCPP (Open Charge Point Protocol) version 1.6 or later. For more info, consult slide 25.

Webfleet EV Smart Charging offers two key features

Charger monitoring

Charger monitoring and dashboard

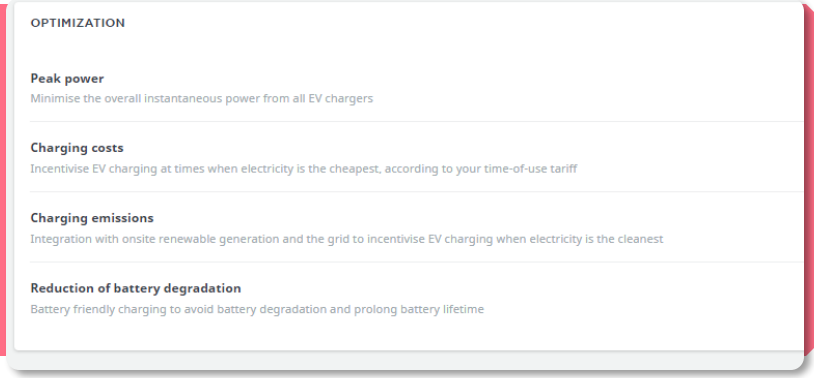
Monitor and manage all charge points on the same platform independent of their brand and conveniently find homogenous data to be on top of the charging expenses[1]



Charger optimisation

Charging optimization considering EV SoC[2]

- Reduce costs by up to 50%
- Reduce peak demand by 80%
- Reduce CO_2 by 70%
- Reduce battery degradation by 40%



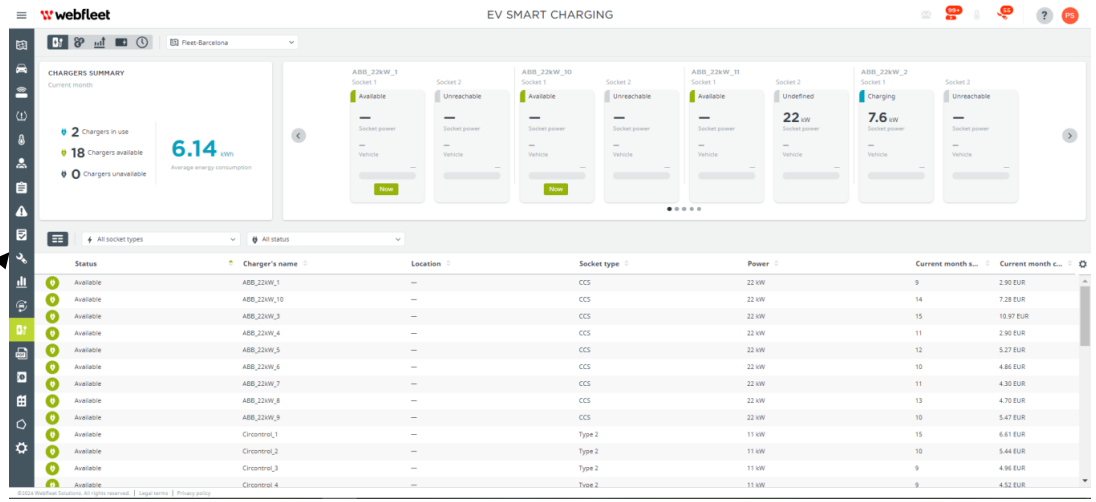
1 – Customer should provide their tariff contract information in order for expenses to be calculated.
2 – Vehicle State of Charge (SoC) considered in planning to avoid compromising fleets' reliability.

Webfleet EV Smart Charging in Webfleet ecosystem

Chargers

Data collected:
- Energy dispensed
- Energy source
- Velocity of charging, charging station availability

Requirements:
- Charger installed at depot[1]
- Charger with internet connection[2]
- Charger supports OCPP V1.6[3]



WEBFLEET UI

Webfleet Telematics Service Platform

Vehicles

Data collected:
- Battery state of charge (SoC)
- Battery levels

Requirements:
- EV model supported by LINK 2, 6 or 7 series[4]
- ECO tariff and above

Business manager

Data collected:
- Energy tariff
- Charger and business routine information[3]
- Departure time

Requirements:
- Energy contract
- Select optimisation criterias

1 – For the moment, only chargers located at same address are monitored. Different facilities monitoring to be released in Q2/2023.
2 – No extra hardware required.
3 – For more info, consult slide 25.
4 – Use the Vehicle Compatibility Checker (VCC) to verify supported vehicles/models.
5 - Order management input to be released in Q1/24.

Smart charging reduces costs, emissions and optimise charging routine

Webfleet EV Smart Charging uses **vehicle data** to safeguard fleet reliability while optimising charging to reduce peak load, energy costs and CO2, and to improve battery lifetime.

-40%

In charging costs

Overpaying by charging during peak hours

-80%

Peak load reduction

Extra fees for exceeding the peak load

-70%

Reduction in CO_2 emissions (estimate)

Cannot charge with renewable sources

-40%

Battery degradation (estimate)

Inefficient charging worsens battery performance

Use cases

Target customers/prospects

- Fleets with **private chargers** installed in the company´s facility (depots, warehouses, company parking)
- Chargers used for commercial fleets
- Vehicle readiness and reliability are key for business operations (e.g. last-mile delivery fleets, service and installation fleets).

Solution	Use case	Proof point	Benefit
EV Charger Monitoring	Fleet manager would like to grow EV fleet, but not sure about utilization of the chargers.	Charger monitoring allow users to monitor the usage and energy required to charge current fleet and see the available energy on site. Optimisation for peak load distributes power equally across existing charge points to peak load	• No need for extra capacity • Possibility to increase fleet of EVs with same available energy capacity
	Fleet manager would like insights into the availability of chargers to better plan vehicle allocation	Charger monitoring allows users to see in real-time charger availability and monitor trends of charger utilisation	• Anticipate planning and react to urgencies (e.g. Malfunction) • Awareness of the impact in planning when chargers are unavailable • Plan orders according to vehicle availability
EV Charger Optimisation	Customer needs to reduce energy costs for charging	Smart charging cost optimisation charges vehicles at off-peak hours	• Reduction in charging costs • Platform allows charging vehicles in off-peak hours
	Customer needs to increase residual value of electric vehicles	Battery care optimisation balances the velocity of charging to increase the lifetime of batteries	• Increased lifetime of batteries • Higher vehicle residual value
	Customer needs to reduce carbon footprint of their operations	Optimisation allows to prioritize energy from renewable sources either on site or coming from the grid	• Carbon savings tracking from charging • Meet carbon savings targets

Overview

EV Charger Monitoring

Charger availability

Monitor in one place the availability and occupancy of all your private chargers.

Vehicle state of charging and charging speed

Get real-time information on the remaining charging time and battery level for vehicles connected to private charging points.

Charger location and charger data

Identify on list or map view the location and description of your chargers.

Charging dashboard

Monitor your charging expenses as well as occupancy rate, average session time and energy consumed in private charging events in a dedicated dashboard to be on top of your charging routine.

EV Charger Optimisation

Peak power optimisation

Avoid consuming the full power capacity or contracting additional capacity to your depot with the peak power (OR load balancing) optimisation.

Charging costs optimisation

Set a target to reduce charging costs and let the platform decide when to charge at lower tariff hours.

Green energy optimisation

Integrate with on-site renewable energy and the grid to incentivize EV charging when electricity is the cleanest.

Battery care

Battery friendly charging to avoid degradation and increase battery lifetime.

EV Charger Monitoring

Charger availability

- Monitor all private chargers in real time using charger monitoring module
- Track the state of your private chargers following the color code:

Charging

Available

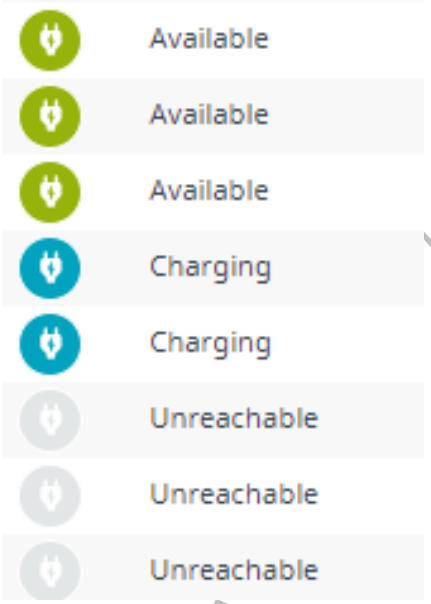
Unreachable/Undefined

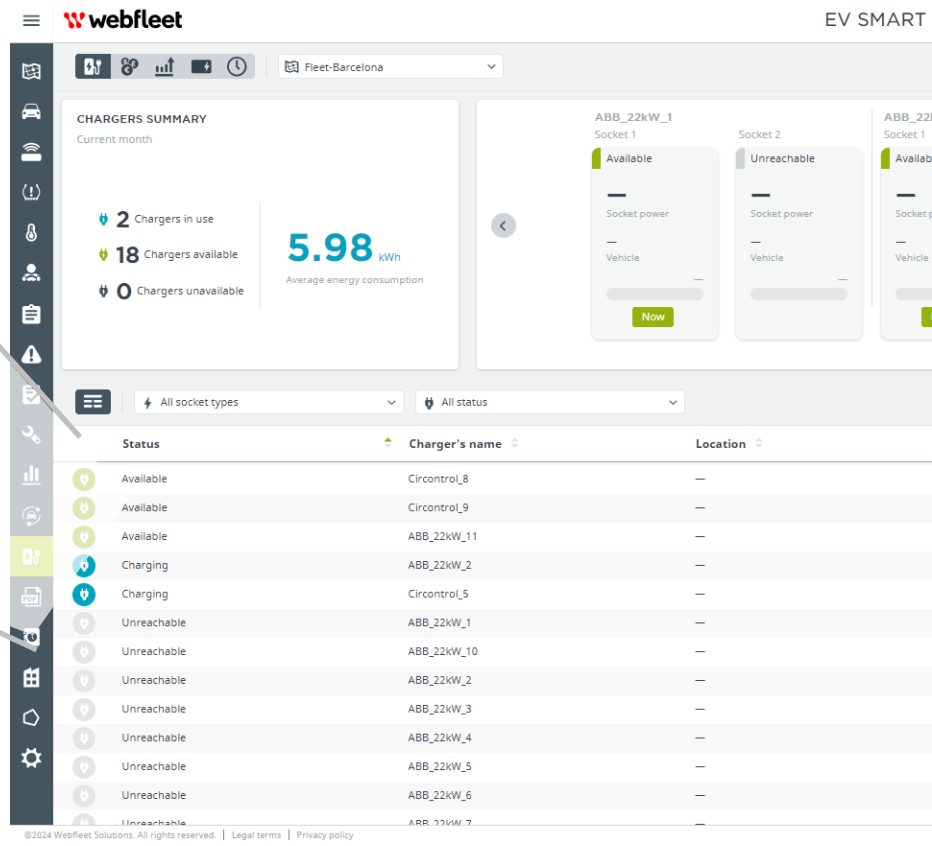
Unavailable

Reserved

Faulted

Preparing/Finishing/Suspended



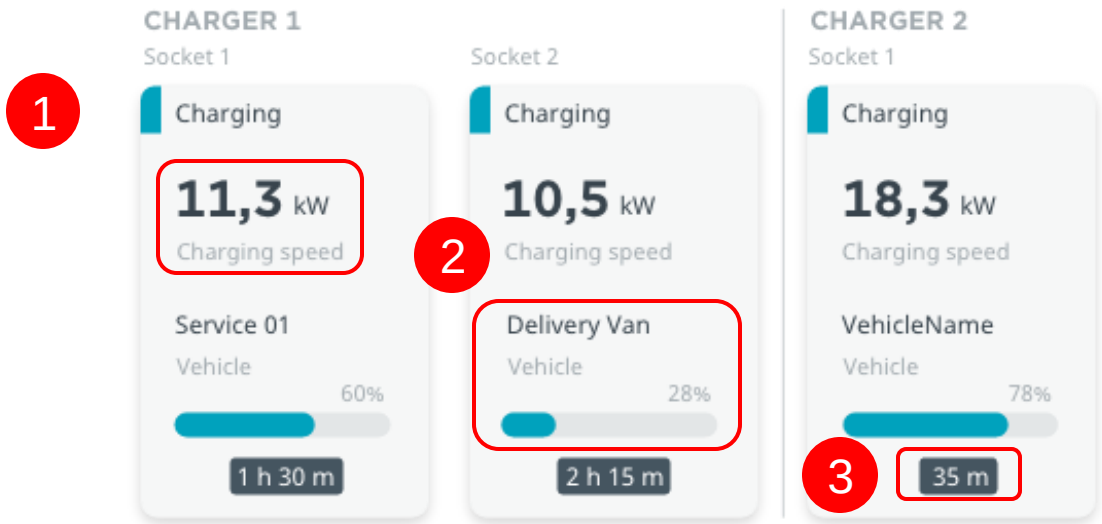


Vehicle state of charging and charging speed

- A blue bar means that a vehicle is connected to that charger. Once charging, the information is provided:

1. Charging power of current session
2. Vehicle ID[1] and current battery level[2]
3. Remaining charging time[2] to reach 100%

- Provided vehicle data allows fleet managers to see in one overview that charging is going well, or if they need to take action to anticipate a longer than expected charging session



1 – Vehicle ID and detailed information will be available in Q3/23
2 – Information is only available for vehicles with LINK 640, LINK 740 or LINK 245 with Webfleet ECO and above

Charger list

- Locate all private chargers connected to Webfleet in the summary list independent of their brand.
- Quickly find information filtering by charger name, availability, facilities and other details.
- A list with charger information can be customised. Default information provides:
  - Charger availability,
  - Charger name
  - Location address
  - Type of socket
  - Power
  - Current month sessions
  - Current month cost

Status	Charger's name	Location
Charging	Charger 1	Lindenplatz 15, 88693 Deggenhausertal, Germany
Charging	Charger 1	Lindenplatz 15, 88693 Deggenhausertal, Germany
Available	Charger 4	Lindenplatz 15, 88693 Deggenhausertal, Germany
Unavailable	Charger 6	Tannenstraße 31, 73099 Adelberg, Germany
Connected	Charger 2	Tannenstraße 31, 73099 Adelberg, Germany
Out of service	Charger 3	Adlzreiterstraße 28, 80337 München, Germany
Unavailable	Charger 6	Adlzreiterstraße 28, 80337 München, Germany
Unavailable	Charger 6	Adlzreiterstraße 28, 80337 München, Germany
Unavailable	Charger 6	Adlzreiterstraße 28, 80337 München, Germany
Unavailable	Charger 6	Adlzreiterstraße 28, 80337 München, Germany

Socket type	Power	Current month sessions	Current month cost
CCS	22 kW	8	2.59 EUR
CCS	22 kW	14	7.28 EUR
CCS	22 kW	12	10.45 EUR
CCS	22 kW	11	2.90 EUR
CCS	22 kW	10	4.82 EUR
CCS	22 kW	10	4.86 EUR
CCS	22 kW	10	4.16 EUR
CCS	22 kW	12	4.26 EUR
CCS	22 kW	10	5.47 EUR
Type 2	11 kW	14	5.10 EUR
Type 2	11 kW	9	4.82 EUR
Type 2	11 kW	8	2.84 EUR

Charger data

- Upon selection of a charger on the list or on the map, it is possible to access charger data:

1 Detailed list and cost per charging session[1]

– Verify in a list all detailed information regarding charging sessions performed

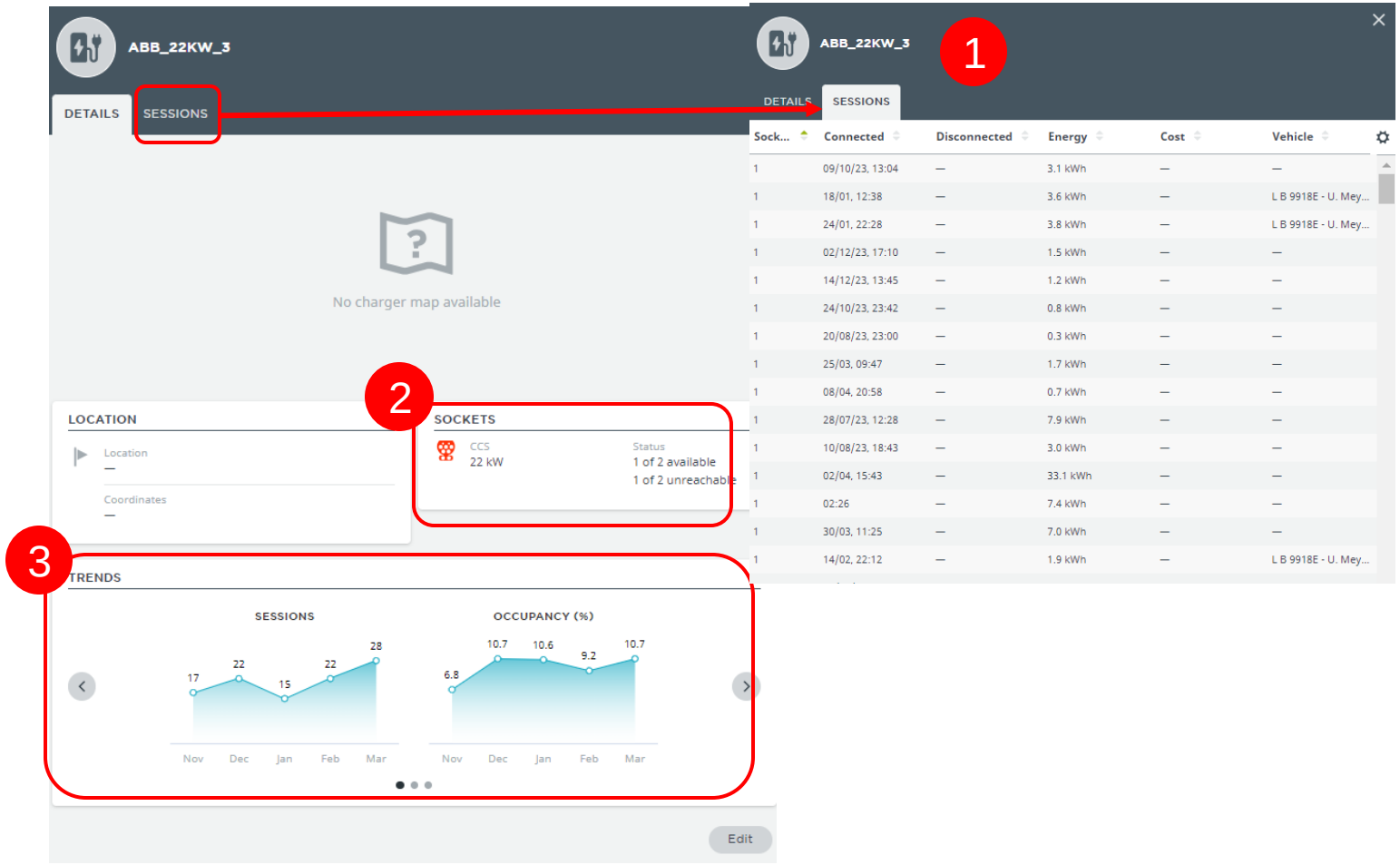
– Identify timestamps of start and end, total energy distributed, vehicle connected, and cost of the session

2 Availability of charging points

Identify the number of available charging points and connector type

3 Trends

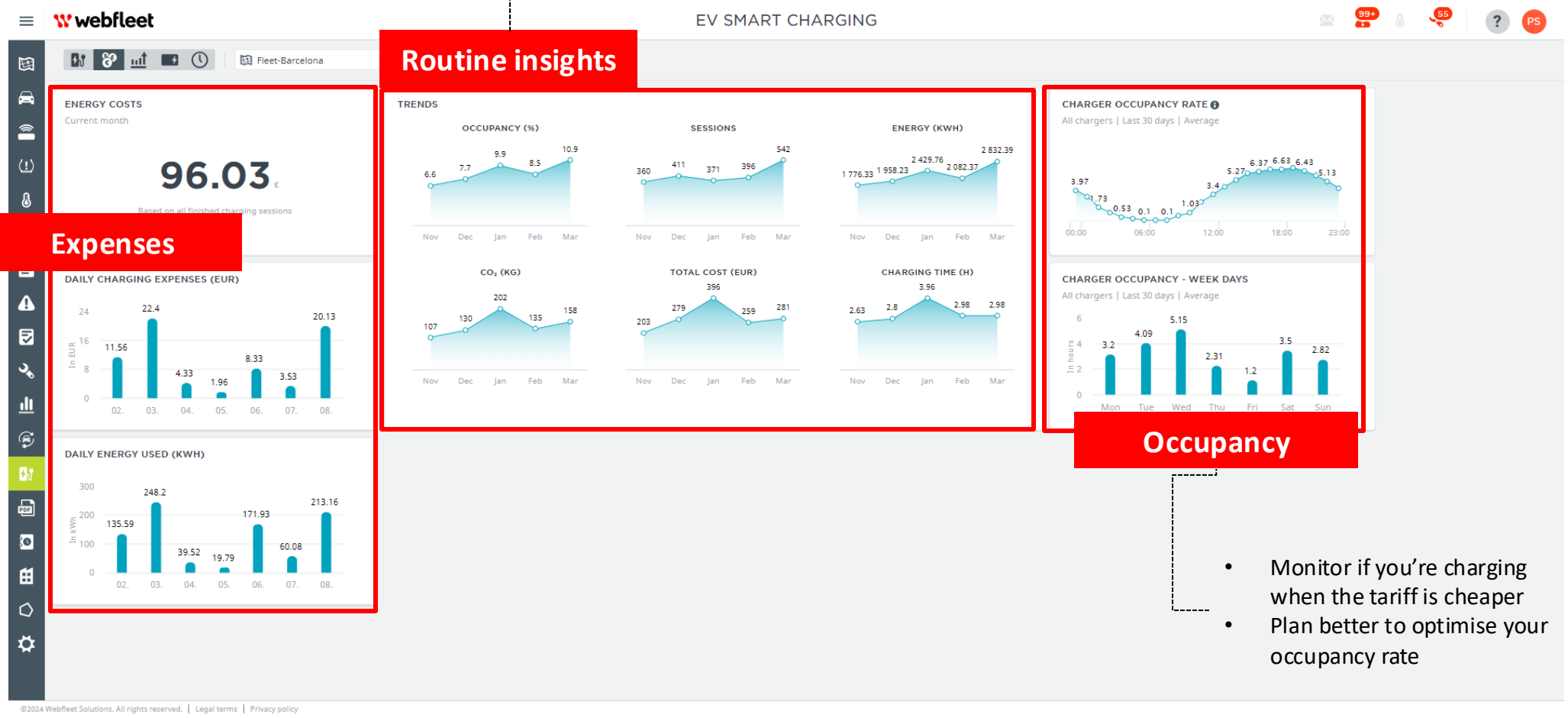
Be on top of the trends in regard to volume of charging sessions and energy provided per charger



1 – Information is only available for vehicles with LINK 640, LINK 740 or LINK 245 with Webfleet ECO and above

Charging dashboard

- Understand average occupancy and idling time of chargers
- Monitor the average charging time and number of sessions performed



- Monitor your total cost of charging
- Track your charging expenses over time
- Monitor energy used for charging

- Monitor if you're charging when the tariff is cheaper
- Plan better to optimise your occupancy rate

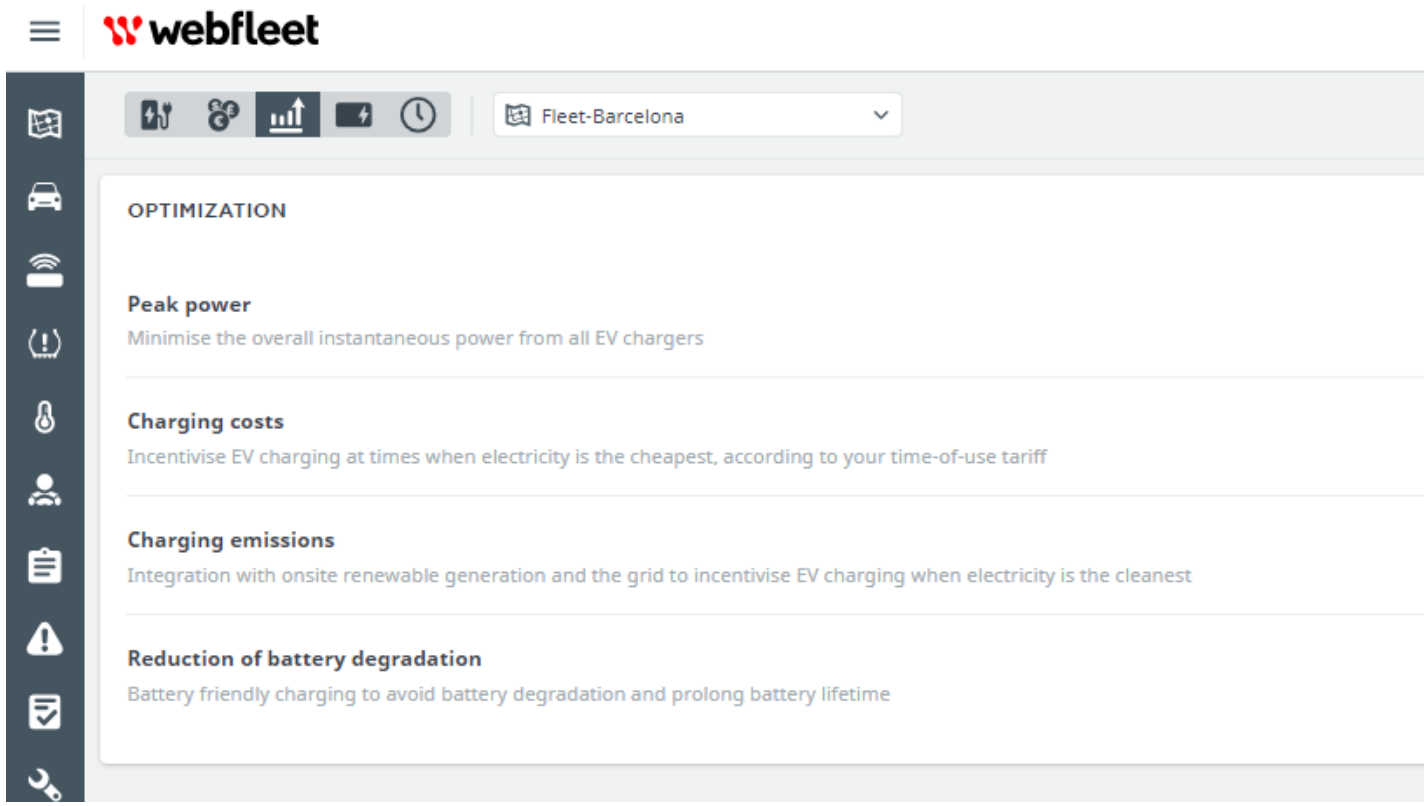
EV Charger Optimisation

What optimisation means on charging routine

In EV's daily charging routine, there are several criteria our Webfleet EV Smart Charging looks at in our optimisation algorithm:

- **Peak power**
  - Avoid peak consumption by charging plugged vehicles at different times and spreading the load
- **Charging costs**
  - Charges vehicles at low energy tariff daily hours
- **Charging emissions**
  - Charges vehicles at hours when energy is coming from renewable sources[1]
- **Reduction of battery degradation**
  - Charges vehicles at lower charging speed and only up to required level to extend battery lifetime

On top of the criteria mentioned above, the charging prioritisation is given to vehicles based on the typical time these vehicles leave and return to the garage.



1 – Available where emission data is provided by the grid operator

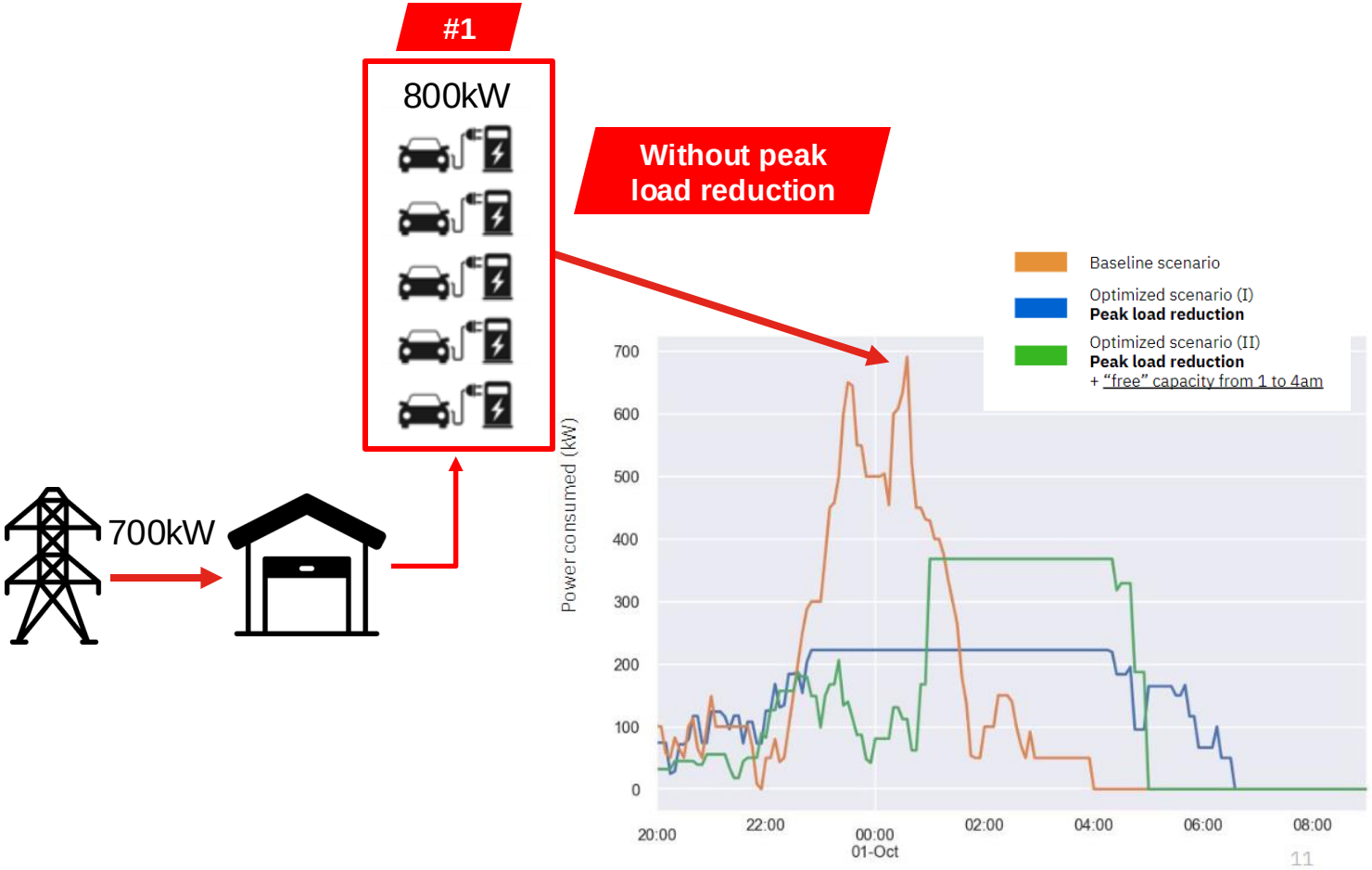
Peak power optimization

Consider the following situation:

- The power grid is able to deliver 700kW to a depot.
- Vehicles return at the end of the day to charge overnight and need to be ready at 8am the following day.

The graph showcases EV loads in the depot across different scenarios:

- **Scenario #1:** without peak load reduction, **vehicles plugged in are charged immediately**. See the depot reaches a very high peak load around 1am.



1 - Order management input to be released in Q1/24.

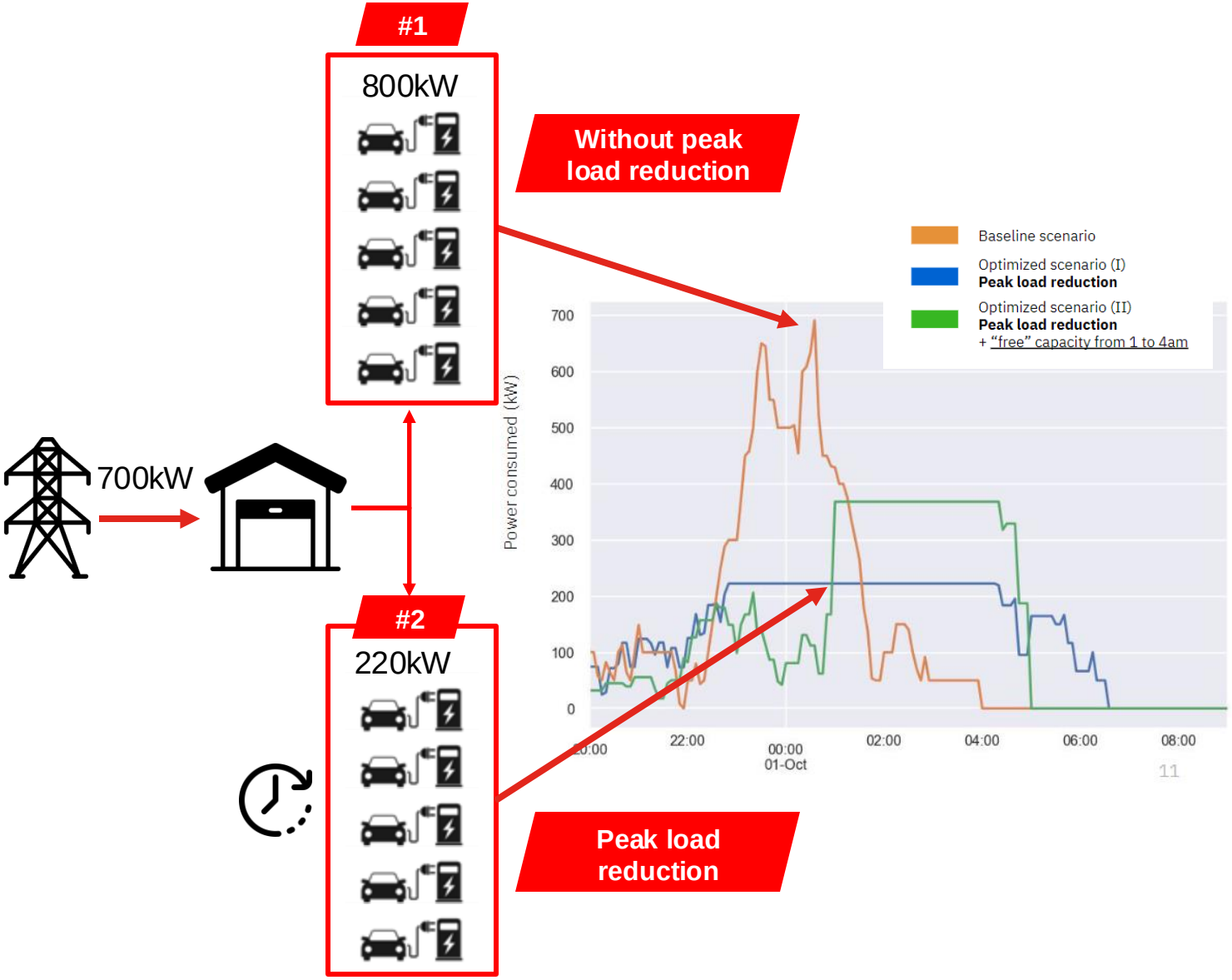
Peak power optimization

Consider the following situation:

- The power grid is able to deliver 700kW to a depot.
- Vehicles return at the end of the day to charge overnight and need to be ready at 8am the following day.

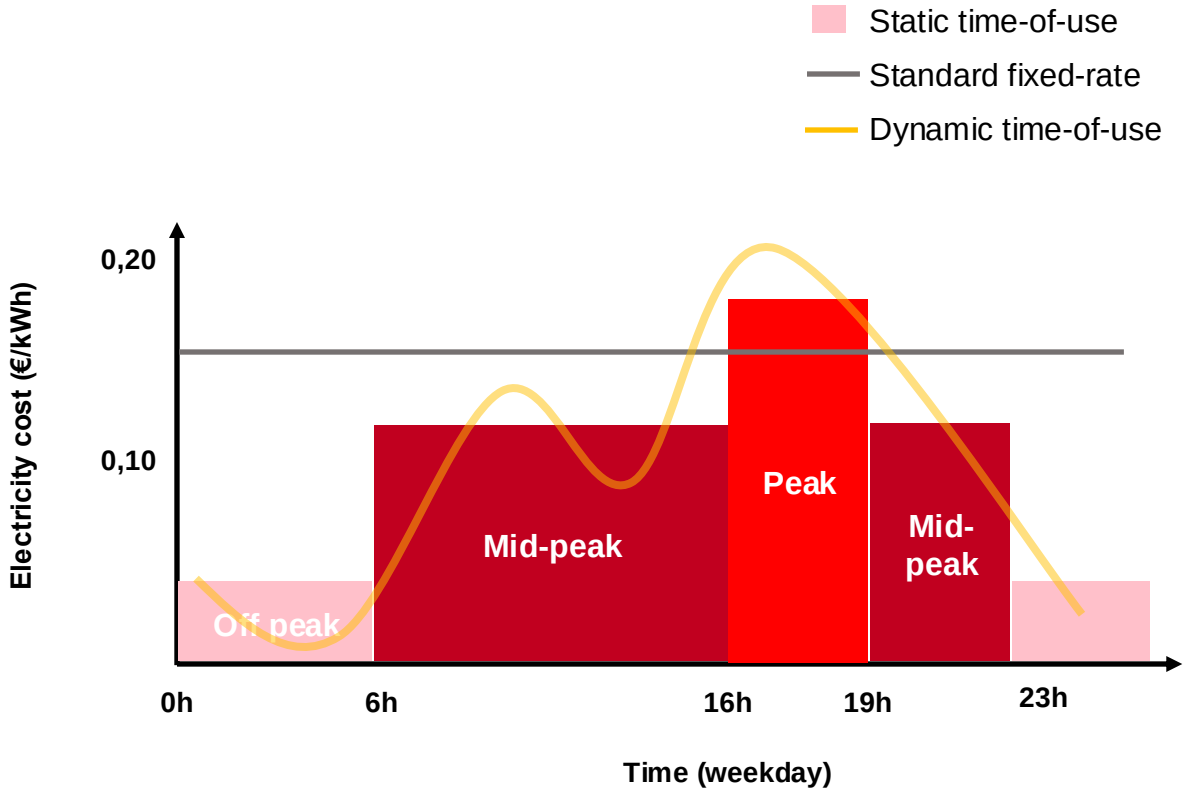
The graph showcases EV loads in the depot across different scenarios:

- **Scenario #1:** without peak load reduction, **vehicles plugged in are charged immediately**. See the depot reaches a very high peak load around 1am.
- **Scenario #2:** with peak power optimization, vehicles are plugged in and the **platform decides when to start charging.** It will **delay** charging of certain vehicles or **charge at lower speed** to balance the energy capacity and avoid peak consumption.



Charging costs optimization

- The platform algorithm will monitor off-peak hours and only start charging vehicles when tariff is at the lowest rate. This optimisation can **provide up to a charging cost reduction of 50%.**
- Energy demand varies along the day. During off-peak hours, energy suppliers encourage increased consumption with cheaper rates.
- Using charging costs, the platform prioritises vehicle charging during off-peak hours.
- To benefit from this optimisation criteria, customer needs to contract variable tariff rate with their energy supplier.
- During onboarding, we will help customer input the contracted energy tariff value to the platform.



Charging emissions optimization

- Enabling charging emissions as optimisation criteria, Webfleet EV Smart Charging prioritises vehicle charging sourced from renewables to **reduce up to 70% of the carbon footprint**.
- If a customer has renewable energy sourcing on site or would like to use green energy from the grid, they can opt for charging emissions.
- Both local renewables production and the power grid emissions factor can be monitored by the Webfleet EV Smart Charging.
- Charging emissions is a key criteria for businesses looking to reduce their carbon footprint and match CO_2 savings targets.

Reduction of battery degradation

- Our estimation is that this criteria can allow a **reduction of up to 40% in battery degradation**.
- The cost of an electric vehicle (EV) battery can account for 30% to 40%[1] of the total EV price. It is the most valuable asset in the vehicle.
- Charging speed can directly influence the extention or reduction in the lifetime of a battery.
- Webfleet EV Smart Charging will care for balancing the charging speed during charging sessions to avoid battery dwelling at high state of charge.

1 – Institute for Energy Research.

Benefits

Lower your running costs

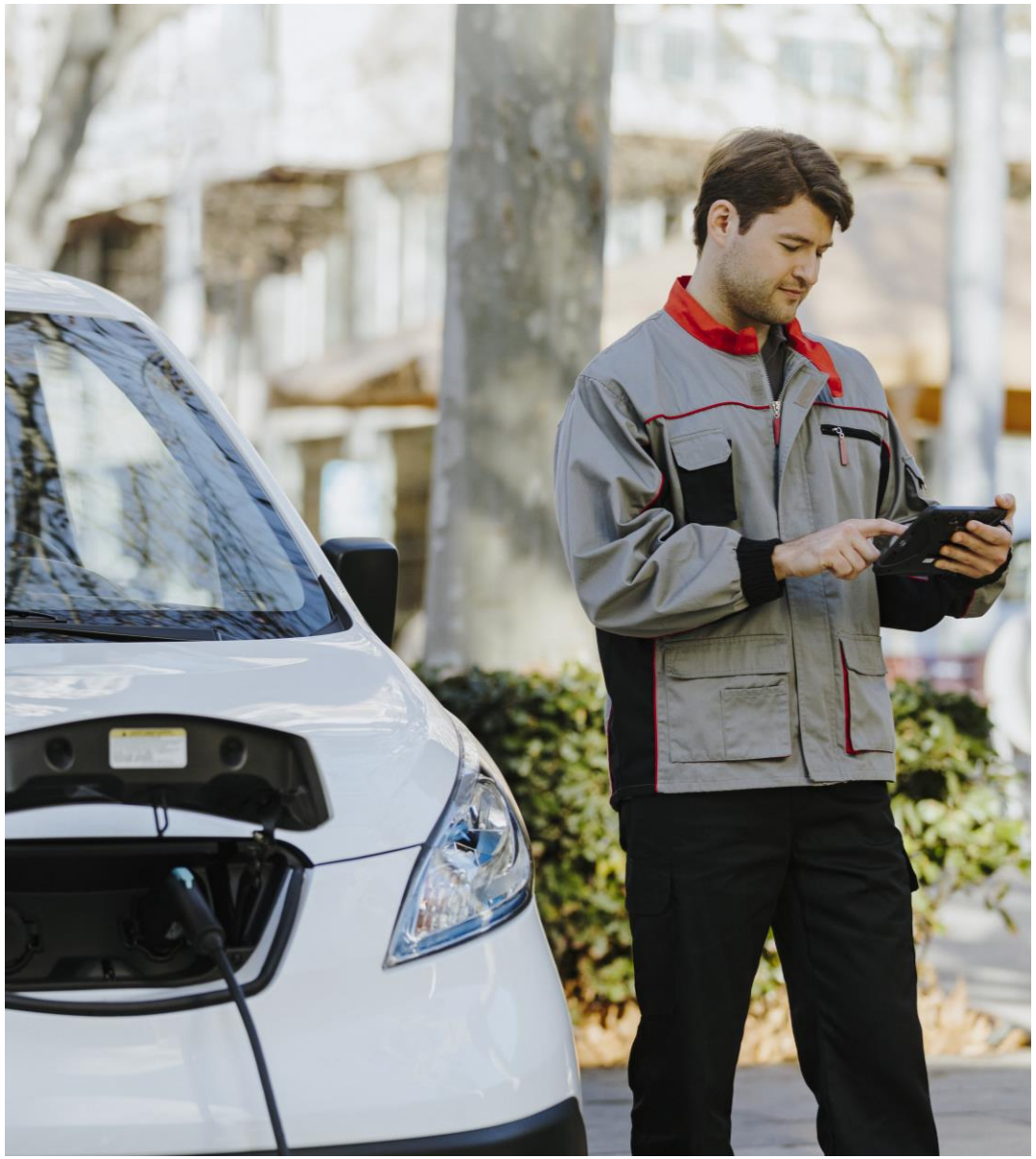
- Save money by optimising available charging capacity and avoiding peak loads
- Charge your EVs with the lowest energy tariffs
- Reduce CAPEX investment for charge points and energy capacity

Manage your fleet and private charging stations on a single platform

- Get an overview of all your chargers across all your locations together with your EVs from your account dashboard
- Connect any compatible charger to Webfleet independent of its brand
- Gather insights on relevant charging metrics and combining information from your charging stations and vehicles

Increase sustainability and prolong the lifetime of your vehicles

- Charge your EVs when renewable energy is most abundant
- Prolong EV lifetime by charging in a battery-friendly way



Requirements and Price

- **Requirements**

  Chargers

  – Chargers must have internet connection

  – Chargers must be OCPP compliant or use a compatible back-office[1]

- **Price and commercial terms**

  Supported sales regions: Europe

  Charger fee – price per socket per charger type (AC or DC); fees are charged per socket per month (consult slide 27 for type of chargers)

  – Charger monitoring package
    - AC charger – 5€ per socket per month
    - DC charger – 10€ per socket per month

  – Charger monitoring + charger optimisation package
    - AC charger – 15€ per socket per month
    - DC charger – 50€ per socket per month

1 – OCPP (Open Charge Point Protocol) allows communication between the charging station and our backend. The charger needs to support at least version 1.6 or later
Compatible back-office requires the provider is a partner on the EV Services Platform. Contact the EV team for more information

Types of chargers

	AC		DC		
Power	up to 7,4kW	7,4kW - 43kW	25kW - 50kW	>100 kW	Up to 150kW
Plug type	Type 1 Type 2	Type 2	CHAdeMO CCS	CHAdeMO CCS	Tesla Type 2 Tesla CCS
Category	Slow	Fast (7-22 kW) Rapid (43 kW)	Rapid	Ultra-rapid	Tesla Superchargers
Typical vehicle type charged	BEVs with small battery packs, PHEVs	BEVs (PV and LCV), PHEVs	BEVs (PV and LCV), PHEVs	BEVs (PV and LCV), e-Buses e-HGVs	Only for Tesla models
Typical location	Home and urban areas	Home and urban areas Business facilities	Business facilities	Motorway services Close to main routes E-buses garages	Tesla supercharger stations
Average price	€ 570 - €1 140	€ 1 140 - €8 500	€ 3 500 - €35 000	€ 25 000 - € 80 000	€ 50 000

Onboarding process – OCPP

1. Qualifying questions

- Are chargers: internet-connected, OCPP 1.6 compliant, owned by customer (not rented/leased)
- Is customer able to reconfigure chargers or has a service provider (CPO) willing to do so?

No?

Large customer? (>20 EVs/chargers)

2. Onboarding form

- Customer indicates scope (monitoring / optimization goals) and provides charger details in onboarding form
- Send filled in form to Webfleet EV team (smartcharging@groups.webfleet.com):

Onboarding form

Webfleet EV Team

3. Charger reconfiguration

- Sales Engineer / EV Consultant / EV Team member confirms all is set up correctly
- Customer (or service provider) reconfigures chargers to new charger monitoring service: ws://ocpp.biapower.io

Questions?

4. Book & Start

- Contract booked via Pyramid per Account Manager or Reseller
- Customer can test if chargers work as expected as soon as data shows up on Webfleet
- After ~2 months check in with customer for feedback

Questions?

Appendix - Acronyms

AC	Alternating Current (typically slow chargers)	EVSE	Electric Vehicle Supply Equipment, a charging unit that can charge 1 vehicle
BEV	Battery Electric Vehicle	FCEV	Fuel-cell Electric vehicle, out of scope as they mainly refuel hydrogen instead of charging
CCS	Combined Charging System, a type of charging plug/socket	kW	KiloWatt, unit for speed of charging
CMS	Charger Management System / Software	kWh	KiloWatt-hour, unit for amount of energy
CPO	Charge Point Operator	(e)MSP	(electric) Mobility Service Provider, supplying payment cards for public charging
DC	Direct Current (typically fast chargers)	OCPP	Open Charge Point Protocol, communication between charger and CMS
(D) LM	(Dynamic) Load Management, preventing peak power demand on grid	OCPI	Open Charge Point Interface, communication between CMS and others
EV	Electric Vehicle; mainly BEVs and PHEVs; PHEVs mostly refuel instead of recharging	RFID	Card or key-fob with a unique ID to authenticate a user/vehicle, to authorize (and sometimes invoice) a charging session

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Solutions for your journey

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